MILITARY SPECIFICATION

LIFE PRESERVER, UNDERARM, CASUALTY, TypE MB-1

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of inflatable life preserver, designated type MB-1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1,1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-1053	Whistle, Ball, plastic.
MIL-S-2785	Shark Repellent Compound, Packet.
MIL-L,-345L	Life Preservers, Packaging of.
MIL-C-5024	Compound, Lubricating, Inner Tube, Aircraft Tire.
MIL-C-5040	Cord , Nylon.
MIL-S-17980	Sea Marker, Packet, Inflatable Survival Equipment..
MIL-C-19002	cloth, Coated , and Tape, Coated Cloth-Polychloroprene on Nylon, Pneumatic Life Preserver.
MIL-C-19377	Cloth, Twill, Nylon.

STANDARDS

FEDERAL

FED-STD-151	Metals, Test Methods.
FED-STD-601	Rubber, Sampling and Testing.
FED-STD-751	Stitches, Seams and Stitchings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4220

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MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of,
MIL-STD-889	Metals , Dissimilar,

DRAWING

AIR FORCE

56K3881	Life Preserver, Underarm Casualty, Type MB-1 Assembly of.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The life preservers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4,4 and 6.4).

3.2 Components. The preserver shall consist of the following major components:

Item No,	Quantity	Item Name	See Requirement
1	1	Flotation cell	3.8.1
2	1	Container	3.8.2
3	1	Emergency signal light	3,8.3
4	1	Whistle	3,8,4
5	1	Sea dye marker	3.8,5
6	1	Shark deterrent	3.8.6
7	1	Lifeline	3,8,7

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

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3.4.1 Elastomeric or rubber materials. Unless otherwise specifically approved by the procuring activity, all elastomeric or rubber materials and component parts used in the life preserver shall have been manufactured within 12 months of the date of delivery to the Government.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to salt spray or atmospheric conditions likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-sTD-889.

3,4.3 Flotation cell cloth. The tape used in the construction of the flotation cells shall conform to type II, III, or IV of MIL-C-19002. The coated cloth used in the construction of the cells shall conform to type I of MIL-C-19002 with the exceptions specified in 3.4.3,1. All defects in the cloth coating not specifically covered by the applicable table of MIL-C-19002 shall be reinforced on the coated side with 1 inch diameter discs of tape. Not more than four such discs shall be permitted in any Cell. The discs shall be applied by cementing.

3.4.3.1 Exceptions to MIL-C-19002. The following exceptions shall apply to the type I coated cloth:

a. Base cloth: The following shall apply in lieu of the sentence that state: that the base cloth shall conform to the requirements of MIL-C-19377: The base cloth shall conform to MIL-C-19377 except that any amendments to MIL-C-19377 shall not apply in lieu of the table entitled “Table of Defects:”

(1) The cloth shall be subjected to quality conformance inspection which shall consist of testing samples from each lot for con formance to section 3 of the specification and an examination of the sample of the filled shipping container from each lot for conformance to the packaging , packing, and marking requirements. A lot shall consist of cloth made essentially under the same conditions and in a single production run. In addition, a sample shall be selected from each lot in accordance with NIL-STD-105 and shall be run over a bank of lights having sufficient intensity to reveal defects in the cloth at a distance of 2 feet. All visible irregularities, including burl marks, shall be considered defects and shall be graded as specified herein,

(2) Defects up to 1/2 inch in any dimension stall be graded one point; defects more than 1/2 inch but not more than 3 inches in any dimension shall be graded two points; defects more than 3 inches but not more than 6 inches in any dimension shall be graded three points; and defects exceeding 6 inches to full width of 1 running yard shall be graded four points. For the purpose of evaluation under MIL-S TD-1OS, the number of points shall be defined as the number of defects, The number of points or defects shall be evaluated using an acceptable quality level of 150 defects per 100 yards.

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(3) Three- and four-point defects shall be flagged in the selvage for removal in the fabrication process. All inperfections protruding from the surface of the- cloth, such as knots, fuzz balls, SlubS, and kinky filling, shall be removed by careful hurling before the cloth is offered for delivery.

b, Resistance to accelerated weathering: Specimens which are tested by the grab method shall have a minimum breaking strength of 80 pounds in the warp direction, 90 pounds in the filling direction, and an ultimate elongation of 20 percent in either direction after weathering. A colorfastness to light equal to the standard color sample shall be maintained after weathering. (Note: Access to the standard color sample may be obtained as specified in 6.3).

c, Acceptable quality, base cloth defects: After the coating process, the number of visible defects that are attributable to the basecloth shall not exceed the number existing in the base cloth and shall not be considered coated fabric defects. Critical coating defects will not be accepted, All coating defects in an accepted roll of finished coated cloth shall be marked on the defect and at the selvage edge adjacent to the defect.

3.4,4 Other materials. All other materials shall conform to the applicable drawings.

3.5 Color. The color of the flotation cells shall be molten orange, and the color of the exposed cement shall blend with the color of the flotation cell fabric. The color of the other components shall conform to the applicable drawings. Variation in the shade of individual materials that is readily apparent shall not be used in one unit.

3.6 Design and construction. The design and constructlon of life life preserver shall conform to drawing 56K3881. The life preserver shall consist of a detachable flotation cell which shall have a mechanical. and oral inflation system. The cell shall be packed in a container that is attached to an adjustable harness, The signal light shall be attached to the cell and the whistle, shark deterrent, sea dye marker, and lifeline shall be attached to the container and assembled as specified in 3.6.1.

3.6.1 Unit assembly. The life preserver shall be assembled as follows : The flotation cell shall be completely deflated by means of a vacuum source through the oral tube and valve assembly. The inflator assemblies with anti-chafing pads and charged carbon-dioxide cylinders shall be mounted on the inflator valve stem of the flotation cell. The emergency signal light shall be attached to the cell by a snap fastener. The *shark* deterrent shall be tied to the grommet on the top left of the container, the whistle shall be tied to the grommet at the top right of the container, and the lanyard of the whistle shall be faked and secured by a rubber band to prevent entanglement during inflation, The lifeline shall be tied to the “I?” ring on the harness, and this remainder shall be faked and secured by masking tape to prevent entanglement during inflation. The components shall be positioned in the container, and the flotation cell shall be folded in a way that will provide an even closure of the container. The inflator lanvards shall be routed through the slot in the container. The fasteners shall be connected at the approximate center of the container, and the closure shall be continued towards the ends of the case. (Note: All preservers shall have charged carbon-dloxide cylinders installed prior to final assembly (see 3.7.4))

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3.6.2 Operation of preserver. When the complete life preserver is properly assembled and donned, a pull on the knobs of the inflation lanyards shall. operate the preserver as follows:

a, Actuate the inflator levers to inflate the flotation cells which separate the fasteners around the container.

b. The shark deterrent, the sea dye marker, and the signal light battery shall drop below the cell and, when in actual service, shall enter the water.

3,6.3 Cementing, Either cementing or pressure-curing by heat shall be used in the construction of seams, sealing areas, and attachments of all component parts of the life preserver. All surfaces to be cemented shall be clean, and all cement used shall be fresh and shall be mixed within 8 hours of application. Fresh cement shall not be put into containers containing congealed cement. The cement shall be applied *over* the total contact areas and shall extend beyond all seam tapes and attachments by at least 1/8 inch on the inside and outside of the cells and by not more than 3/8 inch on the outside of the cells. After cemented areas are jointed, trapped air and wrinkles shall be rolled out. The interior of the cell shall be dusted lightly with compound conforming to MIL-C-5024 to prevent adherence of noncemented surfaces.

3,6.4 Stitching, Unless otherwise indicated on the drawings, the stitches used in the construction of the life preserver shall conform to stitch type 301 of FED-STD-751 with 8 to 10 stitches per inch. All seams shall be backstitched at least 1/2 inch, Stitching shall not be used in the gas-retaining portion of the flotation cells.

3,7 Performance.

3.7.1 Operation. After the life preserver has been properly assembled as specified in 3.6.1, it shall be donned by an individual who shall actuate the cell. The carbon-dioxide cylinders shall be discharged when a reasonable pull is applied to the lanyards. The flotation cell shall inflate, in the time specified, to the following minimum pressures; 0.15 pound per square inch (psi) 20 seconds after discharge of the cylinder and 1.0 psi 45 seconds after discharge of the cylinders. The maximum pressure in the cells shall not exceed 3,0 psi when the temperature of the flotation cells reaches 70 (plus or minus 2) degrees Fahrenheit. The cells shall not leak after this test.

3.7.2 Pressure. The flotation cell shall withstand an inflation pressure of 5 psi for a period of 5 minutes without evidence of failure of the materials or cemented areas and without evidence of permanent distortion of the cell from its designed shape.

3.7,3 Leakage. The flotation cell with components, except the cylinder, shall be inflated to 2 psi. After 24 hours, the pressure in the cells shall be not less than 1 1/2 psi.

3.7.4 Cylinder weight. Before the cylinder is installed on the preserver, the cylinder and the carbon-dioxide charge shall weigh not less than the gross weight marked on the cylinder,

3,7.5 Burst pressure. The flotation cell shall not burst when subjected to a pressure of 12 psi.

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3.7.6 Oral valve attachment. The oral valve shall be attached to the oral tube. The valve shall withstand a load of 50 pounds.

3.7.7 Adhesion. The adhesion of all seams and attachments to the flotation cells shall be not less than 5 pounds per inch of width after joining coated to uncoated sides of the cell cloth and not less than 7 1/2 pounds per inch of width after joining coated to coated sides of the cell cloth.

3.7,8 Leakage after operation. After the life preserver has been inflated 10 times as specified in 4.7.10, the cells shall not leak.

3,7.9 Buoyancy. When the flotation cell of a completely assembled life preserver is inflated by the carbon-dioxide cylinders, the life preserver shall support a weight of 50 pounds in water.

3.7,10 Salt spray. Unless otherwise specified, all metal parts used in the construction of the life preserver shall withstand a 96 hour exposure to salt spray without corrosion of the base metal.

3.8 Details of components.

3.8.1 Flotation cell. The shape and dimensions of the flotation cell shall conform to the applicable drawing. The cell shall have two mechanical inflation systems and an oral inflation system. The cell shall be provided with an internal device or devices to disperse the discharge of gas from the carbon-dioxide cylinders to prevent the cracking of the coating of the flotation cell cloth during inflation, The device or devices shall not abrade nor damage the cloth of the cell. A snap fastener shall be cemented to the upper front of the cell for attachment of the signal light.

3,8.1.1 Mechanical inflation system. Each mechanical inflation system shall conform to the applicable drawing (56K3881).

3,8.1.2 Oral inflation system. The oral inflation system shall conform to the applicable drawing (56K3881).

3,8,2 Container. The container shall conform to the applicable drawing and shall be constructed of one panel, folded in the middle to form an envelope and held closed with snap fasteners.

3.8.2,1 Harness. An adjustable harness shall be attached to the container as shown on the applicable drawing.

3,8.2,2 Container-to-cell attachment. The flotation cell shall be attached to the container by means of snap fasteners,

3.8.3 Emergency signal light. The emergency signal light shall consist of a signal bulb, a water-activated battery, an electric cord, and an attachment patch and shall be attached to the snap fastener assembly on the flotation cell. The emergency signal light used shall be one that has been approved by the procuring activity (see 6.5).

3,8,4 Whistle. The whistle shall conform to MIL-W-1053, type II, and shall be of either split or solid construction,

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3.8,5 Sea dye marker. The sea dye marker shall conform to MIL-S-17980, except that the attaching tape shall be replaced with a 12 inch length of cord conforming to type I of MIL-C-5040. The ends of the cord shall be seared to prevent fraying.

3.8.6 Shark deterrent. The shark deterrent shall conform to MIL-S-2785, except that the attaching tape shall be replaced with a 12 inch length of cord conforming to type I of MIL-C-5040. The ends of the cord shall be seared to prevent fraying

3,8.7 Lifeline. The lifeline shall conform to the applicable drawing and shall be attached to the harness as shown.

3.9 Weight. Excluding the emergency signal light, whistle, sea dye marker, shark deterrent, and lifeline, the complete preserver shall weigh not more than 3.0 pounds. The cell, excluding the mechanical inflator and cylinders, shall weigh not more than 0.70 pound.

3.10 Identification of product. Unless otherwise specified, all markings shall be made with a durable nonmigrating ink of a color that will contrast with the color of the surface that is marked. The ink shall have no deleterious effect on the marked surface. The markings shall conform to MIL-STD-130.

3.10.1 Container. The container shall be marked as follows in 1/4 inch letters and numerals:

Life Preserver, Underarm, Casualty, Type MB-1

Specification MIL-L-25798D(USAF)”

*Manufacturer’ s Identification _____

*Contract or order No. _____

*Date of Manufacture _____

*To be furnished by the Manufacturer,

3.10.2 Flotation cell. The flotation cell shall be marked in 1/4 inch letters and numerals on the upper left front, with the following information:

Cell, Flotation, MB-1 Life Preserver.

*Manufacturer’s Identification”

*contract or order.

*Date of Manufacture.

**serial Number.

*To be furnished by the Manufacturer.

**Serial numbers will be furnished by the procuring activity.

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3. 10.2.1 Location of components. The following information shall be marked in 1/4 inch letters at the location of the applicable component on the upper surface of the flotation cell. One-inch-long arrows shall point to the component.

a. Signal light - to activate pull plug(s) from battery

b. Sea dye marker - pull tab to release

c. Shark chaser - pull tab to release

d. Whistle

e. Oral valve - to increase cell pressure, unscrew knurled knob (counterclockwise) and depress, Immediately blow through tube.

3.11 Workmanship. Workmanship shall be in accordance with high grade manufacturing practices covering this type of equipment. Because of the emergency use of this equipment, all details of workmanship shall be of the highest quality to insure proper functioning of the life preserver at all times.

4, QUALITY ASSURANCE PROVISION

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of life preservers shall be classified as follows:

a. Qualification tests

b. Acceptance tests

c. First article inspection,

4.3 Test conditions,

4,3,1 Atmospheric conditions. Unless otherwise specified, all tests shall be made at an atmospheric pressure of 28 to 32 inches of mercury, a temperature of 70 (plus or minus 18) degrees Fahrenheit, and a relative humidity of 80 percent or less. When tests are concluded with atmospheric pressure or temperature differing from the starting values, corrections shall be made for the change in values.

4,3,2 Temperature corrections. If the ambient temperature changes during the 24-hour leakage test, adjustment shall be made on the basis of a change in pressure of 0,0312 psi per degree Fahrenheit change in temperature.

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4.3.3 Barometric corrections. If the barometric pressure changes during the leakage test, adjustment shall be made on the basis of change in pressure of 0.489 psi per inch (mercury) change in barometric pressure.

4,4 Qualification tests. The qualification test samples (life preservers and materials) shall be subjected to the qualification tests specified in 4,4.2,

4.4.1 Test samples. The qualification test samples shall consist of two life preservers and material samples, (accompanied by test reports) as specified in the letter of authorization.

4.4.2 Tests, Qualification tests shall consist of all tests described under 4.7.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling tests.

4,5.1 Individual tests. Each life preserver shall be subjected to the following tests as described under 4.7:

a. Examination of product.

b, Operation.

c* Pressure.

d, Leakage.

e. Cylinder weight.

4.5,1,1 Rejection. The life preservers may be corrected and submitted to the procuring activity for acceptance, but an air-retaining seam shall not be reworked, under any circumstances , more than one time.

4.5.2 Sampling plans and tests.

4.5,2.1 Lot. A lot shall consist of life preservers manufacturered under essentially the same conditions and submitted for inspection at substantially the same time. A lot shall not exceed 200 life preservers for sampling plans A and B and shall be in accordance with inspection level S-3 of MIL-STD-105 for sampling plan C.

4.5.2,2 Sampling plan A. One life preserver, selected at random from each lot of 200 or fraction thereof, shall be subjected to the following tests as described under 4.7:

a. Burst.

b. Oral valve attachment”

c. Color matching.

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4.5.2.2.1 Rejection, plan A. When one or more items from a lot fail the specified tests, the entire lot shall be rejected. Before resubmitting the lot! the contractor shall fully explain to the Government representative the corrections made and the cause of previous rejections, Life preservers that have been rejected may be replaced or repaired and resubmitted for acceptance, Life preservers rejected after resubmission shall not be resubmitted again without specific approval of the procuring activity, An air-retaining seam shall not be reworked after the life preserver is submitted for acceptance.

4,5.2.3 Sampling plan B. An adhesion test sample shall be prepared from cell fabric by each worker each time a different batch of cement is used by the worker. The sample shall be identified with the lot in production. Five adhesion test samples shall also be prepared from each incoming roll of cell fabric , The cementing procedure shall be the same *as* that used in the construction of the flotation cells. In addition, five sample seams and five sample patches shall be cut from the flotation cell used in sampling plan A. These samples shall be subjected to the adhesion test specified in 4,7.9.

4.5,2.3.1 Rejection, plan B, When one or more of the adhesion test samples specified in 4.5.2,3 fail the specified test, production shall be halted until the cause of failure is determined and the corrections made. When a sample seam from a flotation cell fails the specified test, the lot represented shall be rejected. Lots rejected on the basis of seam-adhesion tests may be resubmitted for acceptance provided five additional flotation cells selected at random from the lot are tested for adhesion and each is found to conform to specified requirements.

4,5,2.4 Sampling plan C. Upon completion of the individual tests specified in 4.5.1. random samples shall be selected from each lot in accordance with inspection level S-3 of MIL-STD-105, The sample size shall be based only on the applicable size code letter corresponding to inspection level S-3. Each life preserver selected as a sample unit shall be forwarded to an inspection activity selected by the procuring activity (see 6.2) for the following tests and examination:

a. Operation.

b, Pressure ,

c, Leakage.

d. Examination of product”

4.5.2.4.1 Rejection, plan C. The acceptance number shall be zero, and the rejection number shall be one for each test and examination. The lot from which the life preserver samples were selected shall not be shipped until the samples have been approved by the procuring activity and the lot has been accepted by the cognizant quality control representative. The lot from which the life preserver sample was selected shall be rejected if any sample fails to meet any requirement of the tests specified in 4.5.2.4, A rejected lot shall not be resubmitted except with the approval of the procuring activity. Upon completion of the tests and examinations, the sample will be returned to the manufacturer.

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4,6 First article inspection, Unless otherwise specified (see 6.2), the samples specified in 4,6.1 shall be submitted as specified in 4.6.1.1 for the performance of examinations and tests deemed necessary by the testing activity.

4,6.1 Samples. First article samples shall consist of three complete life preservers of the type specified in the contractor order. The samples shall be representative of the construction, components , materials, and workmanship that will be used during production.

4.6.1.1 Submittal of samples, Unless otherwise specified, as soon as practicable after the award of the contract or order, samples specified in 4,6,1 shall be forwarded as specified in the contract or order, When a contractor is in continuous production of life preservers from contract to contract, submission of further samples on the new contract may be waived at the discretion of the procuring activity. Approval of the first article samples or the waiving of the first article inspection does not preclude performance of the acceptance tests.

4,6,1.2 Identification of samples, The first article samples submitted shall be plainly identified by securely attached tags marked with the following information: Sample submitted by (name) (date) for first article inspection in accordance with MIL-L-25798D(USAF) under contract number _____

4.6.1.3 Disposition of samples, Disposition of the first article samples shall be as specified by the procuring activity (see 6.2).

4,6.1.4 Results of inspections. Results of inspections, and where applicable, recommendations and comments pertinent for use in monitoring production shall be kept available for the procuring activity.

4,7 Test methods.

4,7.1 Examination of product, The life preserver shall be visually examined to determine compliance with the requirements of this specification with respect to workmanship, materials, marking, and construction.

4.7.2 Operation. This life preserver, assembled as specified in 3.6,1, shall be adjusted to an individual to simulate actual service use, The carbon-dioxide cylinders shall be discharged simultaneously with a reasonable pull to determine if the cell will come out of the container (see 3,6.2). The components shall not be tangled or in any way prohibit the cell from inflating to its design shape, The minimum internal cell pressure 20 seconds after discharge of the cylinders shall be 0,15 psi. The minimum internal cell pressure 45 seconds after discharge of the cylinders shall be 1.0 psi. The maximum internal cell pressure shall not exceed 3.0 psi when the cell temperature has reached 70 (plus or minus 2) degrees Fahrenheit, Failure of the cells to inflate as specified or evidence of leakage or failure of the mechanical inflation assembly shall be cause for rejection. After this test is completed, the flotation cell shall be completely deflated by means of a vacuum source and the life preservers shall be subjected to the test specified in 4.7,3.

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4.7.3 Pressure. The cell of the life preserver shall be inflated to 5 psi with oil-free air. After a minimum of 10 minutes, the pressure shall be checked and readjusted, if necessary, to the original pressure of 5 psi. The temperature and pressure shall be recorded at this time. A minimum of 5 minutes after the readjustment period, the pressure shall be measured and corrected for any change in temperature or barometric pressure. The corrected pressure shall be not less than 4.60 psi, After these measurements and corrections , the pressure shall be released. Any evidence of failure of the material of cemented areas or any evidence of permanent distortion of the cell shall be cause for rejection.

4.7.4 Leakage. The carbon-dioxide cylinders shall not be attached for this test, The cell of the life preserver shall be inflated with oil-free air to a pressure of 2 psi. After a minimum of 15 minutes, the pressure shall be checked and readjusted, if necessary, to the original pressure of 2 psi, The pressure source shall then be removed. A minimum of 24 hours after the readjustment period, the pressure shall be measured and corrected for any change in temperature or barometric pressure. The corrected pressure shall be not less than 1 1/2 psi. *There* shall be no leakage from the valve stem. This test shall follow the test specified in 4.7.3. Failure to meet this test shall be cause for rejection.

4.7,5 Cylinder weight. Each cylinder shall be weighed not more than 72 hours prior to its installation on the life preserver, The weight of the cylinders and carbon-dioxide charge shall be not less than the gross weight marked on the cylinder,

4.7.6 Burst. The flotation cell shall be subjected to an inflation pressure sufficient to cause failure of material or seam areas. The pressure at which failure occurs and the type of failure shall be recorded, Failure at a pressure of 12 psi or less shall be cause for rejection.

4.7.7 Oral valve attachment, The strength of attachment of the oral valve in the tube shall be determined by applying force along the longitudinal axis of the assembly in a tensile machine with a jaw separation of 12 inches (plus or minus 1/2 inch) per minute, Failure to withstand a load of 50 pounds shall be cause for rejection.

4.7,8 Color matching. The color of the cell shall be compared to and shall approximately match the color of the standard shade (see 6.3). The color of the container shall conform to the applicable drawing. Individual materials in which the color variation is apparent shall not be used together in the same unit .

4.7.9 Adhesion. Adhesion shall be determined in accordance with method 8011 of FED-STD-601.

4.7.10 Leakage after operation. The life preserver shall be subjected to 10 operations as specified in 4,7,2, At the conclusion of these operations, the internal cell pressure shall be adjusted to 2 1/2 psi and the cells submerged in water for not less than 2 minutes. Any evidence of leakage of the cell or inflation assemblies shall be cause for rejection, Any nonconformance to the pressure requirements of 4.7.2 during the 10 operations shall be cause for rejection.

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4.7.11 Buoyancy. A completely assembled life preserver shall be inflated with carbon-dioxide cylinders , The discharged cylinders shall be left in the inflators . A 25 pound steel weight shall be tied to each side of the flotation cell by a 2 foot-length of cord so that the 50 pounds is distributed evenly. The weighted life preserver shall be placed in water and shall not submerge. Failure of the life preserver to support the weights as specified shall be cause for rejection.

4.7.12 Salt spray, Each metal part of the assembly not protected against corrosion by salt spray in accordance with Government specifications, standards , or drawings shall be subjected to a 96 hour salt spray test in accordance with method 811 of FED-STD-151. After this test, each metal part or assembly shall be examined for corroded areas. A corroded area, which is defined as corrosion of the base metal, shall be cause for rejection of the part of assembly.

5. PACKAGING

5.1 Preservation, packaging, packing and marking of the life preservers shall be in accordance with A, B or C of MIL-P-3454, as specified (see 6.2).

5.2 Marking. In addition to any special markings required by the contract, order , or MTL-P-3454, the unit intermediate, and exterior containers shall be marked as follows: LIFE PRESERVER, UNDERARN, CASUALTY, TYPE MB-1.

6. NOTES

6.1 Intended use. The type MB-1 life preserver covered by this specification is intended to be used as emergency flotation gear by adult casualties aboard aircraft of the Military Airlift Command during flights over ‘water.

6.2 Ordering data.

6,2,1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Selection of inspection activity for sampling plan (see 4.5.2.4) and first article samples (see 4.6.1.1).

c. Disposition of samples (see 4.6.1.3 and 6.6).

d. Request for results of first article inspection (see 4.6.1.4).

e. Selection of applicable levels of preservation and packaging and packing.

6.3 Standard color sample. Requests for access to a standard color sample for the cell should be addressed to the: SA-ALC/MMIPCC, Kelly Air Force Base, TX 78241.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether *or* not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Engineering Division, SA-ALC/MMEDO, Kelly AFB, Texas 78241 and information pertaining to qualification of products may be obtained from that activity.

6.5 Acceptable sources for signal light. The following manufacturer's part numbers will be" considered acceptable for use as the emergency signal light (see 3.8.3). Use of a signal light other than those listed herein will require the approval of the procuring activity.

a. Part No, L-8-1 (ACR Electronics Corp., 551 W. 22nd St. , New York 11, New York).

b, Part No. L-8 (McMurdo Instruments Co., Limited, Ashtead, Surrey, England).

c. Part No. EW-104-WL-12-12 (Eureka–Williams Corp. , Bloomington, Illinois).

6.6 Disposition of samples. The disposition of the samples should be as follows: One approved life preserver will be returned to the manufacturer for use in monitoring production; a second approved life preserver will be returned to the manufacturer along with the final sample as required by sampling plan C (see 4.5.2.4); and the third life preserver will be consumed or destroyed in the first article inspection.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions , modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in the notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 4220-F294)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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